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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

Novembre 2025

Note

[illegible]



SPEECH-TO-TEXT

Note

[illegible]

- Simone Scannapieco

Note

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```

...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-starter-model-openai</artifactId>
</dependency>
...

```

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
```

Note

[illegible]

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

Note

[illegible]

- 1 Creazione interfaccia ed implementazione del servizio
- 2 Modifica controllore MVC
- 3 Test delle funzionalità con Postman/Insomnia

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```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

}
```

Note

[illegible]


```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;

    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile);

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content());

    }

    ...
}
```

[illegible]

```
...
private String getContentType(Resource multimediaFile) {
    String contentType = URLConnection.guessContentTypeFromName(multimediaFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = multimediaFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                default -> "media/unknown";
            };
        } else {
            contentType = "media/unknown";
        }
    }

    return contentType;
}
}
```

Note

[illegible]

```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@Configuration
public class QuestionController {

    ...

    private final MultiModalityService multiModalityService;
    ...

    public QuestionController(QuestionService service,
        RAGService ragService,
        TimeToolsService timeToolsService,
        MultiModalityService multiModalityService,
        ...) {

        ...
        this.multiModalityService = multiModalityService;
        ...

    }

    ...

    @PostMapping("/gemini/ask/multi-modality/transcribe")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {
        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }

}
```

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Branch:

16-spring-ai-gemini-ollama-multimodality-audio-transcription

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